

## Vibrational Geometrodynamics (VG):

- A Bridge Between Loop Quantum Gravity and String Theory –

### Abstract

Is there an age limit for daring to propose new theories in physics ? I ask because I am 75 years old, and yet I dare to present this Unifying Vibrational Model – a bridge between giants.

This paper proposes an unifying framework , Vibrational Geometrodynamics (VG), that synthesizes the core insights of Loop Quantum Gravity (LQG) and String Theory (ST). The central thesis is that the spin networks of LQG constitute the fundamental resonant medium of spacetime , where the onedimensional vibrational modes propagating along their links are identified with the fundamental strings of ST.

Consequently , strings are not conceived as vibrating within a pre-existing spacetime; rather, their vibrational dynamics are postulated to constitute spacetime itself. Within this framework , the perceived 3+1 dimensional classical spacetime and its material content emerge as stable interference patterns – holographic projections – of these underlying vibrations on the dynamic membrane of the spin network.

### A personal Note:

This work represents a deep, symbiotic collaboration between a 75-year-old physicist and an AI system. The intuition, vision and persistence are human – the systematization , referencing and drafting were AI -facilitated. Perhaps this too is pioneering – not just the theory , but how it was born.